https://syedaliqasim.github.io/

Syed Ali Qasim

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EDUCATION

Virginia Commonwealth University **Richmond, VA** **Aug 2018 – July 2023**

- Ph.D. in Computer Science
- Advisor: Dr. Irfan Ahmad
- Relevant Coursework: System Security and Forensics, Database And Application Security, Natural Language Processing, Cloud Computing, Distributed Systems, Network Penetration Testing, and Defense, Introduction to Computer Forensics, Advance Machine Learning

University of New Orleans **New Orleans, LA** **2017 – 2018**

- Coursework towards Ph.D. (Transfer).

Lahore University of Management Sciences **Lahore, Pakistan** **2012 – 2016**

- B.S. in Computer Science, 2016

RESEARCH AND PROFESSIONAL EXPERIENCE

Assistant Professor **Grand Valley State University** **Aug 2023- Present**

- Assistant Professor of Computer Science, College of Computing
- Director FORCE lab (Forensics and Offensive Research in Cyber-physical Environments)

Research Assistant **Safe lab VCU** **2018 – July 2023**

- Worked with Dr. Irfan Ahmad on various projects related to computer and industrial control systems security and forensics.
- Developed a tool (PREE) for reverse engineering of ICS protocols.
- Developed a tool (Reditus) for forensic analysis of network-based attacks on ICS
- Developing a virtual PLC framework for ICS Threat Intelligence.

Research Assistant **CyPhy Lab UNO** **2017 – 2018**

- Studied the problem of designing scalable forensic tools for ICS.
- Designed and developed a virtual controller for forensic analysis of Allen-Bradley PLCs.

RESEARCH INTERESTS

- Cybersecurity
- Digital Forensics
- Industrial Control Systems
- Industrial Internet of Things
- Threat Intelligence

RESEARCH FUNDING

Funded

- **R1- CSCE Catalyst Grant - GVSU** "Developing a Smart Fuzzing Tool for Industrial Control Systems" Role: Sole PI, Funding: $7,608, Project Duration: 1 year (May 2025 - May 2026)
- **R2- SPGA - GVSU** "KALAM: Discovering Vulnerabilities in Industrial Control Systems Protocols Using Smart Fuzzing" Role: Sole PI, Funding: $14000, Project Duration: 1 year (August 2025 - May 2026)

Pending

- **R3- NSF-ERI** "ERI: Model-Guided Characterization of Embedded Communication Interfaces in Wireless Medical Sensing and Actuation Systems " Role: PI, Funding: $200000, Project Duration: 2 years. Submitted [Oct,2026]

SELECTED PUBLICATIONS

- **[ICCPS]. Syed Ali Qasim**, Muhammad Taqi Raza, Irfan Ahmed, "PLCpot: Application Dialogue Replay based Scalable PLC Honeypot for Industrial Control Systems", In the 16th ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS), May 2025, Irvine, California, USA
  (Acceptance rate (27%)
- **[ICSS]. Syed Ali Qasim**, Muhammad Taqi Raza, Irfan Ahmed, "vPLC: A scalable PLC testbed for IIoT security research", In Ninth Annual Industrial Control System Security (ICSS) Workshop, December 2023, Austin, Texas, USA
- **[DFRWS USA]. Syed Ali Qasim**, Wooyeon Jo, Irfan Ahmed, "PREE: Heuristic Builder for Reverse Engineering of Network Protocols in Industrial Control Systems", In the 23rd Annual Digital Forensics Research Conference (DFRWS USA'23), July 2023, Baltimore, Maryland, USA
  (Acceptance rate (28%): 16 out of 56 submissions)
  (BEST STUDENT PAPER AWARD)
- **[IEEE Security and Privacy Journal].** Adeen Ayub, Wooyeon Jo, **Syed Ali Qasim**, Irfan Ahmed, "How are industrial control systems insecure by design? A deeper insight into real-world PLCs", In IEEE Security and Privacy Journal, 2023
  (Impact Factor: 3.1 in 2022-23)
- **[IEEE Access].** Masrik Dahir, **Syed Ali Qasim**, Irfan Ahmed, "Cronus: An Automated Feedback Tool for Concept Maps", In IEEE Access Journal, August 2021
  (Impact Factor: 3.75 in 2021)
- **[ICCIP]. Syed Ali Qasim**, Adeen Ayub, Jordan A Johnson, Irfan Ahmed, "Attacking the IEC-61131 Logic Engine in Programmable Logic Controllers in Industrial Control Systems", In the 15th IFIP International Conference on Critical Infrastructure Protection (ICCIP), March 2021, Arlington, Virginia. (held virtually)
  (Resulted in a CVE: CVE-2023-2310)
- **[DFRWS USA]. Syed Ali Qasim**, Jared Smith, Irfan Ahmed, " Control Logic Forensics Framework using Built-in Decompiler of Engineering Software in Industrial Control Systems", In the 20th Annual Digital Forensics Research Conference (DFRWS'20), July 2020, Memphis, TN. (held virtually)
  (Acceptance rate (40%): 12 out of 30 submissions)
  (BEST STUDENT PAPER AWARD)
- **[ISC]. Syed Ali Qasim**, Juan Lopez Jr, Irfan Ahmed, "Automated Reconstruction of Control Logic for Programmable Logic Controller Forensics", In the 22nd Information Security Conference (ISC'19), September 2019, New York.
  (Acceptance rate (26.7%): 23 out of 86 submissions)
- **[CHI EA]. Syed Ali Qasim**, Harris Durani, Muhammad Naeem, Waleed Riaz, Suleman Shahid, "Supporting Pakistani Farmers Through Digital Means: An Exploratory Study", In the proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems (CHI EA '16)

PEER-REVIEWED PRESENTATIONS, DEMOS, AND POSTERS

- **[HotSoS 2023]. Syed Ali Qasim**, Jared Smith, Irfan Ahmed. "Control Logic Forensics Framework using Built-in Decompiler of Engineering Software in Industrial Control Systems", In the 10th Annual Event of The Hot Topics in the Science of Security (HotSoS) Symposium, Virtually hosted by the NSA, April 2023. (Presentation)
- **[CCI]. Syed Ali Qasim**, Irfan Ahmed, "Industrial Control Systems Threat Intelligence Using Virtual PLC Platform", In Commonwealth Cyber Initiative (CCI) Symposium 2023, Richmond, Virginia, USA, April 2023. (Poster)
- **[CINA]. Syed Ali Qasim**, Adeen Ayub, Muhammad Haris Rais, Irfan Ahmed, "Digital Forensic Tools and Techniques for Investigating Control Logic Attacks in Industrial Control Systems", In The Criminal Investigations and Network Analysis (CINA) Annual Meeting, George Mason University, Fairfax, Virginia, USA, October 2022. (Poster)
- **[AAFS]. Syed Ali Qasim**, Irfan Ahmed, "A Case Study of Virtual Programmable Logic Controllers (PLC) Forensic Framework on Investigating Control Logic Attacks in a Belt Conveyor System", In the 74th Annual Meeting of the American Academy of Forensic Sciences, February 2022, Seattle, Washington. (Extended Abstract)
- **[ACNS CIMSS]. Syed Ali Qasim**, Irfan Ahmed, "Virtual PLC-assisted Forensic Analysis of Control Logic Injection Attacks", In the 1st International Workshop on Critical Infrastructure and Manufacturing System Se-

curity (CIMSS) in conjunction with the 19th International Conference on Applied Cryptography and Network Security, Kamakura, Japan, June 2021. (Demo)

TEACHING EXPERIENCE

Instructor **Grand Valley State University**

- Fall 2023
  - CIS 162 – Computer Science I
  - CIS 430 – Computer and Cyber Forensics
- Winter 2024
  - CIS 338 – Wide Area Network Engineering
  - CIS 430 – Computer and Cyber Forensics
  - CIS 617 - Digital Forensics & Investigation
- Summer 2024
  - CIS 617 - Digital Forensics & Investigation
- Fall 2024
  - CIS 418 – Secure Software Engineering
  - CIS 458 – System Security
- Winter 2025
  - CIS 338 – Wide Area Network Engineering
  - CIS 430 – Computer and Cyber Forensics
  - CIS 615 - Information Security Principles
- Fall 2025
  - CIS 456 – Reverse Engineering and Malware Analysis
  - CIS 430 – Computer and Cyber Forensics
  - CIS 615 - Information Security Principles
- Winter 2026
  - CIS 456 – Reverse Engineering and Malware Analysis
  - CIS 617 - Digital Forensics & Investigation
  - CIS 615 - Information Security Principles

Teaching Assistant **Virginia Commonwealth University** **Spring 2020**

- CMSC 414: Computer and Network Security

Teaching Assistant **Lahore University of Management Sciences** **Fall 2015**

- CS 100: Computational Problem Solving

STUDENT RESEARCH ADVISING

Masrik Dahir	$BSComputerScience, VCU \rightarrow$Costar Group	Jan 2021- August 2021
Fiyinfoluwa Olajide	MS Computer Science, GVSU	August 2024- December 2024
Nitin Shrestha	MS Computer Science, GVSU	August 2025- December 2025

SOFTWARE/HARDWARE VULNERABILITY DISCLOSURE

- CVE-2023-2310, Schweitzer Engineering Laboratories (SEL) RTAC 3505, https://selinc.com/support/security-notifications/external-reports/ RTAC 3505

ACADEMIC SERVICES

Program Committee

- ICSS 2023, 2024, 2025
- DFRWS 2025, 2026

Reviewer

- DFRWS USA 2022,2023,2024,2025, 2026
- CIMSS 2021
- DFRWS EU 2023
- ICSS 2023

HONORS AND AWARDS

- **Best Student Paper Award**
  23rd Annual Digital Forensics Research Conference (DFRWS USA 2023), Baltimore USA
- **2022 CCI Innovation Award**
  Commonwealth Cyber Initiative Symposium, 2023 USA
- **Best Student Paper Award**
  20th Annual Digital Forensics Research Conference (DFRWS USA 2020), Held Virtualy
- **LUMS Tuition Fee Waiver Award**
  100% tuition fee waiver for four years, 2012-16

COMMERCIALIZATION AND PATENTS

- **Company Name:** VirtualPLC LLC
  Co-Founders: Irfan Ahmed and Syed Ali Qasim
  Product: Virtual programmable logic controllers (PLCs) for threat intelligence
  Seed Funding: Virginia Commonwealth Cyber Initiative (CCI)

Skills and Technologies

- **Programming:** Python, Java, C++, MySQL, C (In order of expertise).
- **Tools Platforms:** Hadoop, CUDA, Matlab, SQL, Autopsy, Rapid Minor, Weka.